Capital Allocation I

How much risk, and whose

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NYU Stern — Data-Driven Investing

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Today

In the slides

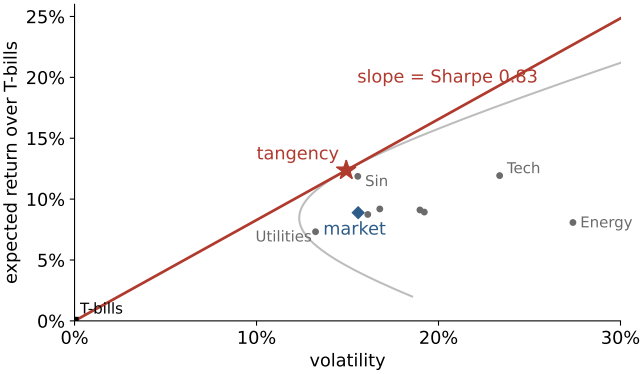
- The frontier, and the formula behind it
- When a new asset moves the frontier: the alpha test
- How your job and your mandate tilt the portfolio

In the notebook, you will

- Build the tangency and minimum-variance portfolios on six factors, step by step
- Draw the frontier from those two
- Size one portfolio for different goals
- Run the alpha test on momentum and on HML

Homework: goals and mandates on the 49 industries

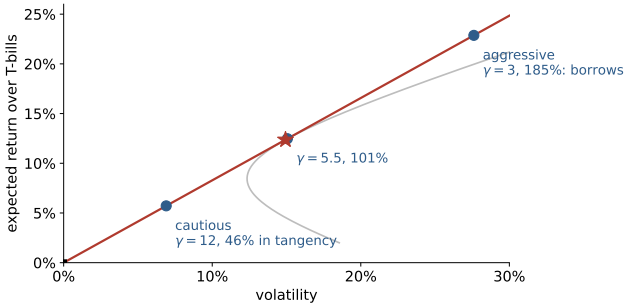
Everyone wants to be on the line



- Each dot is a sector, 1980–2026
- Mix the sectors and you can reach the curve
- Add T-bills, and you can reach any point on the straight line
- The steepest line wins. Its slope is the highest Sharpe ratio: 0.83. The market: 0.57.

Below the line, you could have earned more for the same risk.

Two decisions, and they separate



1. How do you spread it?

The tangency portfolio. The same for everyone.

2. How much risk?

Where you sit on the line. Different for everyone.

A cautious investor does not hold safer assets. They hold less of the same portfolio.

That clean split rests on three assumptions

the assumption	when it fails	what changes
All your wealth is in financial markets	You have a job, a business, a house	Your portfolio should tilt
You choose for yourself	It is someone else's money	A mandate bends the line
You know μ and Σ	You estimated them	*Capital Allocation III*

First the formula. Then we break the first two, with examples.

One asset, no matrices

You put a fraction x of your money in one risky asset, the rest in T-bills.

$$\max_x \; x\mu \; - \; \frac{\gamma}{2} x^2 \sigma^2$$

μ: expected excess return σ: its volatility γ: how much you dislike variance

Differentiate and set to zero: $\mu - \gamma x \sigma^2 = 0$

$$x^\star \; = \; \frac{1}{\gamma} \, \frac{\mu}{\sigma^2}$$

- More return, hold more. More *variance*, hold less, and it goes with the square.

The volatility you run is your Sharpe ratio over your risk aversion

Multiply both sides by σ: $$x^\star \sigma = \frac{1}{\gamma}\frac{\mu}{\sigma} = \frac{SR}{\gamma}$$

γ	weight in the market	volatility you run	
1	3.66	57%	nobody does this
2	1.83	29%	
5	0.73	11%	about three-quarters in stocks
10	0.37	6%	

The market, 1980–2026: 8.9% a year over T-bills, 15.6% volatility, Sharpe ratio 0.57

- The formula answers the question you asked. It will recommend 3.7 times leverage if you hand it $\gamma = 1$.

Volatility management is this formula with σ moving

$$x_t^\star = \frac{1}{\gamma}\frac{\mu_t}{\sigma_t^2}$$

UMD, by last month's volatility	calm	2	3	4	volatile
average return (% a year)	10.7	12.4	6.8	7.2	-0.3
volatility (% a year)	6.5	9.1	11.1	16.5	28.3

- When volatility rises, the mean does not rise with it. So μ/σ^2 falls, and you should hold less.
- Dividing by last month's variance was $x^\star$, with a guess that μ stays put. Same signal, same stocks: Sharpe ratio $0.45 \to 0.94$.

We come back to this in *Conditional Strategies*.

Many assets: the same three steps

Now W and μ are vectors, and Σ is the covariance matrix.

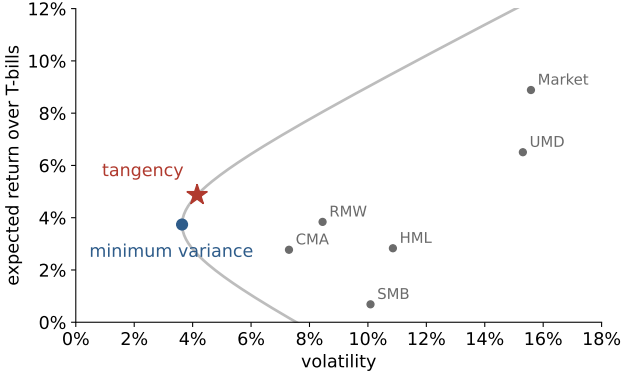
$$\max_W \; W'\mu \; - \; \frac{\gamma}{2} W'\Sigma W \qquad \Longrightarrow \qquad \mu - \gamma \Sigma W = 0$$

$$W^\star \; = \; \frac{1}{\gamma} \Sigma^{-1} \mu$$

- The inverse is where the division used to be
- This is the mean-variance efficient portfolio, the tangency portfolio. It has the highest Sharpe ratio there is.
- It needs two things: μ and Σ. Remember that.

Three ways to get here

Two portfolios draw the whole curve



Six factors, 1980–2026. Each scaled so the weights add to one:

Tangency: $\Sigma^{-1}\mu$

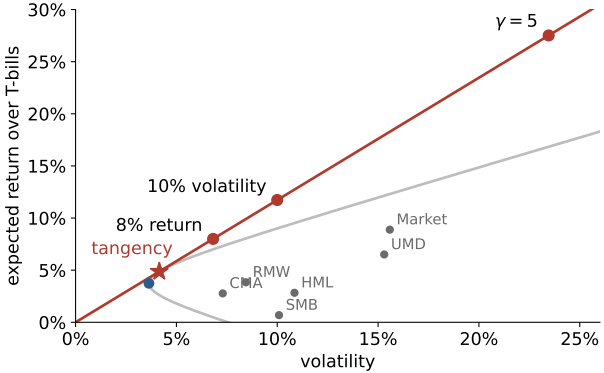
highest Sharpe ratio, 1.17

Minimum variance: $\Sigma^{-1}\mathbf{1}$

lowest risk, 3.6%. It never looks at μ.

- Every point on the curve is a mix of these two
- Both beat every factor by far: diversification

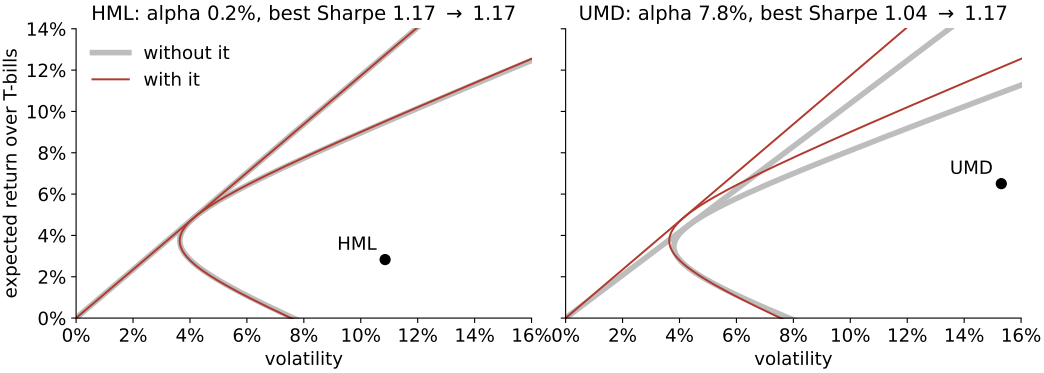
One portfolio, any size



the goal	times tangency	vol
8% return	1.6	6.8%
10% volatility	2.4	10.0%
$\gamma = 5$	5.6	23.5%
$\gamma = 1$	28.2	117%

- Same mix, same Sharpe ratio, 1.17. Only the size changes.
- $\gamma = 1$ is off the chart. The formula answers the question you asked.

Does a new asset move the frontier?



- HML has a Sharpe ratio of 0.26. Take it away and the frontier does not move.
- Momentum has 0.43, not so different. Take it away and the best Sharpe ratio falls by 0.13.
- So a good Sharpe ratio on its own is not what moves the frontier. What is?

The alpha test is a test of the frontier

Regress the new asset on the tangency portfolio of what you already hold:

$$R_{\text{new}} = \alpha + \beta\, R_{\text{tangency}} + \varepsilon \qquad SR^2_{\text{with}} = SR^2_{\text{without}} + \left(\frac{\alpha}{\sigma_\varepsilon}\right)^2$$

add this factor to the other five	Sharpe alone	α a year	t	best Sharpe without	with
Market	0.57	12.7%	6.1	0.74	1.17
RMW	0.45	4.1%	3.6	1.03	1.17
UMD	0.43	7.8%	3.5	1.04	1.17
CMA	0.38	2.2%	2.8	1.09	1.17
HML	0.26	0.2%	0.2	1.17	1.17
SMB	0.07	1.2%	0.9	1.17	1.17

$\alpha = 0$ means the tangency portfolio already spans the new asset. You have run this test all semester.

Why the α is the same

Now: the notebook

Notebook: Capital Allocation I — build the frontier, one step at a time

1. **Tangency, then minimum variance.** Six factors, line by line.
2. **Draw the frontier** from those two portfolios.
3. **One portfolio, many goals.** γ, a return target, a volatility target, all on one picture.
4. **Add a factor.** Run the alpha test on momentum, then on HML.

Bring back: why does HML, with a positive Sharpe ratio, add nothing? Would you drop it?

Stop at **Back to the slides**; the challenge after it is homework.

What the notebook showed

- **The mix and the size are separate.** Four goals, one tangency portfolio, one Sharpe ratio.
- **Assets earn their place through alpha.** HML: positive Sharpe ratio, no alpha, no effect. The other factors already span it.
- **That does not mean drop value forever.** Its alpha is $0.2\% \pm 1.2\%$, in this sample, gross of costs. *Capital Allocation III* is about exactly this.

Everything so far assumed the same tangency portfolio is right for everyone. Is it?

From here we go back to the sectors: tilts and mandates need things with names.

How much risk? First: whose money is it?

The principal: your own money

- How much do you mind losing?
- When do you need it?
- What are you saving for?
- What else do you own?

The delegate: someone else's money

- A volatility target
- A stop-loss, a point where you are fired
- A tracking-error budget
- Limits on what you can hold

Both are answers to one question: where on the line?

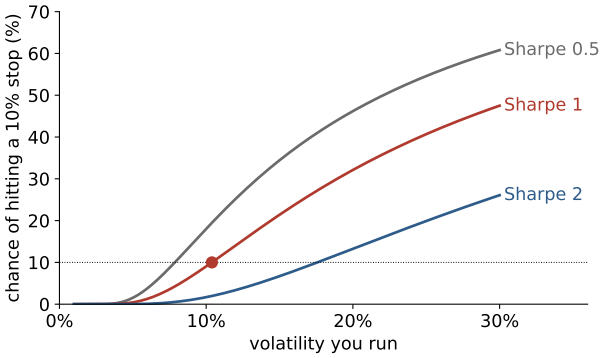
The principal: what is the worst you could stand?

The market fell 50% from 2007 to 2009. If it happens again, how much could you stand to lose?
Nobody knows their γ. Everybody can answer this.

the worst you could stand	so hold in the market	volatility you run	your γ
10%	20%	3%	18
20%	40%	6%	9
30%	60%	9%	6
50%	100%	16%	3.7

- Answer it before the fall. Selling at the bottom is the expensive way to learn your γ.
- Same market, four people: each row is a different point on the same line.

The delegate: a stop-loss is a limit on volatility



- A pod shop might halve a team's capital after a 5% loss, and close it at 7.5%
- The stop is in percent, so volatility decides how often you hit it
- Sharpe 1, 10% stop: at 10% volatility, 9% a year. At 20% volatility, 32%.
- A better strategy can run more risk

The formula

A stop sets your place on the line

your stop	5%	7.5%	10%	20%
volatility for a 10% chance of hitting it in a year, Sharpe 1	5%	8%	10%	21%
the γ that volatility implies, SR/σ	19	13	10	5

- Rule of thumb, at Sharpe 1: run a volatility about equal to your stop
- The principal cannot give you their γ. A stop gives you one.
- Why they need it: a share of the gains and no share of the losses rewards volatility
- *Hiring and Firing* fired on the appraisal ratio, which leverage cannot change. A stop in percent can be managed by running less risk. That is the point, and the cost.

Your biggest asset is not in the market

	financial wealth, W	human capital, H	$h = H/W$
at 22, starting at \$100k	\$10k	\$2.3m	230
at 45, earning \$150k	\$500k	\$2.2m	4
at 65, retiring	\$2m	\$0	0

Human capital: wages to 62, growing 2% a year, discounted at 5%. Stylized.

- Human capital is an asset, and you cannot sell it
- It has a beta. It goes up and down with some things and not others.
- If it is most of your wealth, ignoring it means choosing most of your portfolio by accident

Your job is a position

Choose the financial account so that account plus job is the portfolio you want:

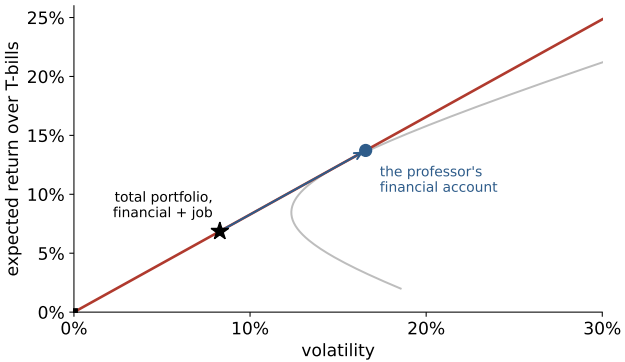
$$w = \underbrace{(1+h)\frac{1}{\gamma}\Sigma^{-1}\mu}_{\text{the same tangency portfolio, sized for total wealth}} - \underbrace{h\,b}_{\text{short what looks like your job}}$$

$h = H/W$ b: slopes from regressing your job's return on asset returns μ, Σ: as before

- Your job already holds b. So the account holds less of it.
- With no job, $h = 0$, and it is the formula from before
- Merton (1971), Bodie, Merton and Samuelson (1992)

Derivation

A safe job: the same mix, more risk



- A tenured professor: the job pays like a bond, $b = 0$
- They already own a lot of "T-bills". So the account takes the risk.
- Same mix, further along the line: here 17% volatility, not 8%
- The job runs out, h falls, and the account moves back

$\gamma = 10$, $H = W$

Now a job that is not safe

Same wealth, same risk aversion, same job security.

One is a software engineer: AI makes them more valuable.

One is a legal translator: AI may replace them.

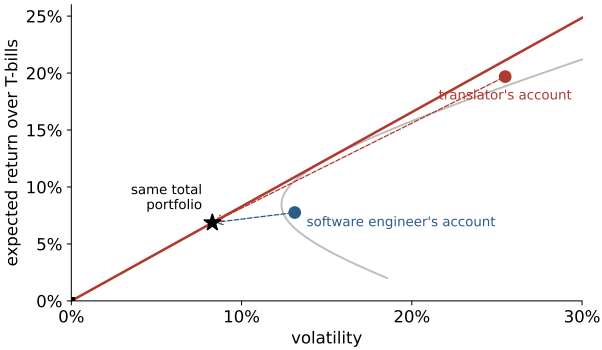
Who should hold more tech stocks?

A The engineer

B The translator

C The same

AI: two accounts, one portfolio



$H = W$, $\gamma = 10$. Job's return: $\pm 0.5\times$ Tech.

	Tech in account
engineer	-11%
safe job	$+39\%$
translator	$+89\%$

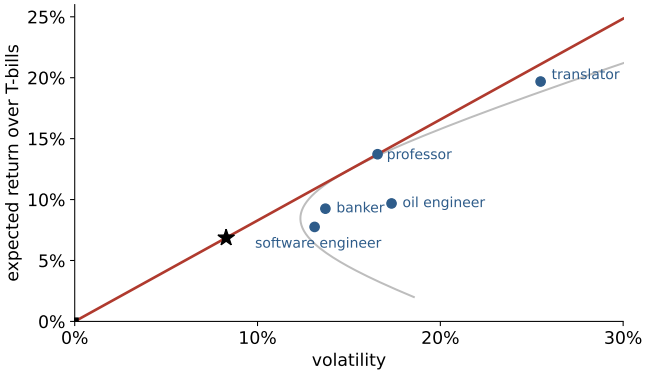
- The translator is buying insurance: AI wins, the job loses, the account pays
- Add the job back and both are at the star

Think it through: what does it move with, and which way do you tilt?

who	what it moves with	the account
tenured professor	nothing much, like a bond	same mix, more risk
investment banker	the market and Finance; the bonus falls with them	less Finance, less market
oil engineer in Houston	Energy, and so does the house	less Energy
software engineer	Tech, and AI	less Tech
translator, paralegal	against AI	more Tech
small-business owner	their sector, and one firm	less of that sector, less risk overall
homeowner	house prices in one city	less real estate
retiree	no job, $h = 0$	the account is everything: less risk
pension fund	owes long-dated payments: a *short* bond	holds long bonds, to hedge what it owes

Now you: what will your job move with?

All of them on one picture



- Same γ, same wealth, same star
- The safe job stays on the line. Every other account falls below it.
- Account Sharpe ratios: 0.59 to 0.77, against 0.83

Below the line, on purpose. Judge the account alone and it looks like a mistake.

What people actually do is the opposite

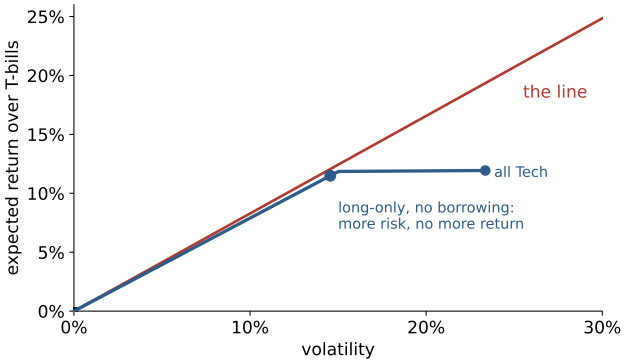
- **Enron, 2001.** About 60% of its 401(k) assets were in Enron stock. When it failed, employees lost the job and the savings together.
- **Familiarity.** Investors tilt *toward* stocks close to their own income, not away from them (Massa and Simonov 2006)
- **Back to the poll.** The instinct is to hold what you know, so the engineer holds Tech.

What you know best is often what you already own too much of

The limits of this

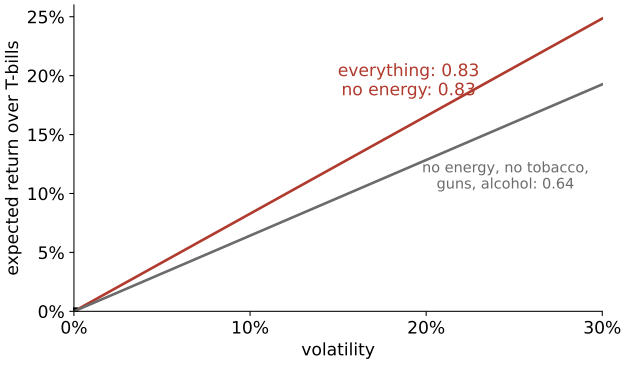
- **You cannot borrow against your job.** At 22 the formula wants many times your savings in the market. You cannot borrow that, so this limit is the one that binds.
- **Nobody can estimate b.** Think sign and rough size, not decimals.
- **Tech is not AI.** Which stocks win from AI is itself a forecast, and insurance has a price: the translator's account earns a lower Sharpe ratio.
- **Most job risk cannot be hedged.** One firm, one boss. What you cannot trade is background risk, and it lowers how much market risk you want (Heaton and Lucas 2000).

A mandate bends the line: no shorting, no borrowing



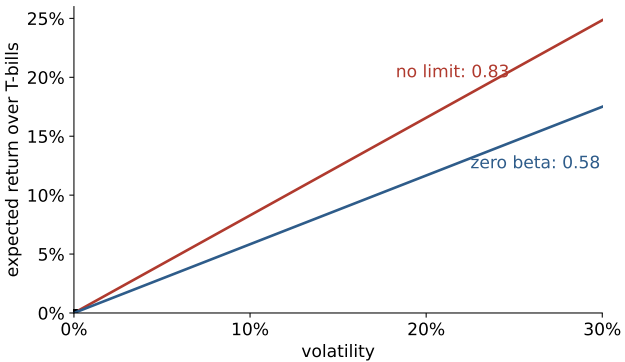
- No shorting barely costs here: 0.83 to 0.79
- No borrowing is what bites. At 18% volatility: 11.9%, not 14.9%.
- To take more risk you must hold riskier assets, not more of the tangency
- Many do, and that may be why high-beta assets earn too little (Black 1972; Frazzini and Pedersen 2014)

Exclusions and ESG: the price depends on what you exclude



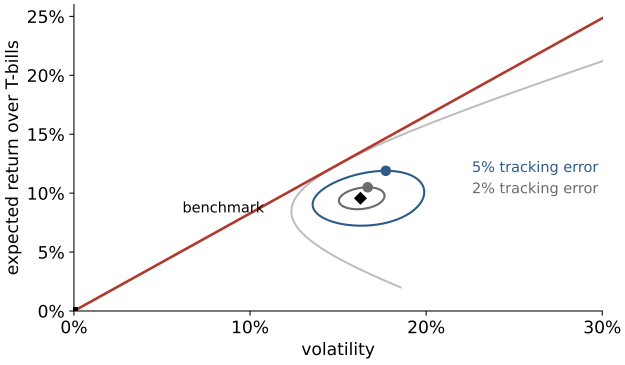
- No energy: 0.83 to 0.83. It cost nothing here.
- No tobacco, guns, alcohol: 0.83 to 0.64. The tangency was 94% in them.
- Excluded stocks may earn more *because* they are excluded (Hong and Kacperczyk 2009)
- Accepting a lower Sharpe ratio for what you value is a preference, not an error (Pedersen, Fitzgibbons and Pomorski 2021)

Factor limits: zero beta takes one leg off the triangle



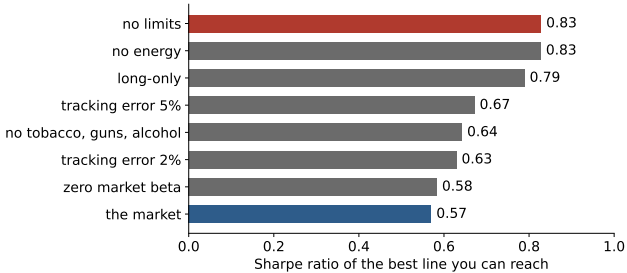
- Market-neutral: the best you can do with no market exposure is 0.58
- Why accept it? The principal already owns the market. They pay you for what they do not have.
- Limits on size, value or momentum work the same way: each removes something the optimizer wanted
- The hedged book is next class

Tracking error: a budget around a benchmark



- You are judged on the gap to the benchmark, so your budget is a small circle around it
- The best point on it is still below the line: Sharpe 0.63 at 2%, 0.67 at 5%
- Reaching the line at the benchmark's risk takes about 12% tracking error
- It also runs more total risk than the benchmark (Roll 1992). The manager does not count that; the owner does.

Every mandate has a price: the slope you lose



- Measure the price first, then decide whether it is worth paying
- A mandate can be worth it: the owner has other holdings, values, and limited trust
- All of this is in sample, and tobacco did very well. In *Capital Allocation III* a constraint can even *help*, because the inputs are estimated.

Takeaways

1. Everyone wants the steepest line. How much risk is a separate decision, if all your wealth is financial.
2. The volatility you run is your Sharpe ratio over your risk aversion. For a delegate, a stop sets it.
3. Your job is a position. Hold the same tangency portfolio, minus what looks like your job.
4. Every mandate has a price: the slope you lose. Measure it, then decide.

Homework: goals and mandates. On the 49 industries: hit a volatility goal, a return goal and a leverage limit. Then take out more and more of the sin industries. What does it cost, and what would happen to their prices if everyone did it?

Appendix

Why the alpha is the same both ways

Take $W = \Sigma^{-1}\mu$. Its variance equals its mean: $W'\Sigma W = W'\mu = \mu'\Sigma^{-1}\mu$. Let c be the covariances of the new asset with the old ones.

- **On the tangency portfolio:** $\beta = c'W/W'\Sigma W$, so $\alpha = \mu_{\text{new}} - \beta\, W'\mu = \mu_{\text{new}} - c'\Sigma^{-1}\mu$
- **On all the assets:** $b = \Sigma^{-1}c$, so $\alpha = \mu_{\text{new}} - b'\mu = \mu_{\text{new}} - c'\Sigma^{-1}\mu$
- The same α. The standard deviation is not the same: in the Sharpe formula, σ_ε is the residual on *all* the assets, which hedges more than the tangency portfolio alone.
- In the table, momentum's α is 7.84% both ways. Scaling W changes β, not α.

Back

A: Three problems, one portfolio

maximize utility	$\max_W \; W'\mu - \frac{\gamma}{2} W'\Sigma W$	$W = \frac{1}{\gamma}\Sigma^{-1}\mu$
hit a return with the least variance	$\min_W \; W'\Sigma W$ s.t. $W'\mu = \bar{\mu}$	$W = \frac{\bar{\mu}}{\mu'\Sigma^{-1}\mu}\Sigma^{-1}\mu$
take a risk budget, earn the most	$\max_W \; W'\mu$ s.t. $W'\Sigma W = \bar{\sigma}^2$	$W = \frac{\bar{\sigma}}{\sqrt{\mu'\Sigma^{-1}\mu}}\Sigma^{-1}\mu$

- All three are a number times $\Sigma^{-1}\mu$. Only the size changes.
- The Sharpe ratio is the same in all three: $\sqrt{\mu'\Sigma^{-1}\mu}$.
- A delegate with a volatility target solves the third, and never needs γ.

Back

A: The chance of hitting a stop

Treat the account as earning $\mu = SR\,\sigma$ a year with volatility σ, watched continuously. The chance of being down D from the start at some point within T years:

$$P \;=\; \Phi\left(\frac{-D-\mu T}{\sigma\sqrt{T}}\right) \;+\; e^{-2\mu D/\sigma^2}\,\Phi\left(\frac{-D+\mu T}{\sigma\sqrt{T}}\right)$$

- This counts losses from the start of the year. A stop on the drawdown from the peak is hit more often.
- Returns are assumed normal, and volatility constant. Real crashes jump through a stop.
- Monthly checks instead of continuous ones would lower the chance a little

Back

A: Why your job is a position

Per dollar of total wealth, you hold x in the assets and $\phi = H/(W+H)$ in your job, whose return is r_H:

$$\max_x \; x'\mu + \phi\,\mu_H - \frac{\gamma}{2}\left(x'\Sigma x + 2\phi\, x'\sigma_H + \phi^2\sigma_H^2\right) \quad \Rightarrow \quad x = \frac{1}{\gamma}\Sigma^{-1}\mu - \phi\,\Sigma^{-1}\sigma_H$$

σ_H: the covariances between your job and the assets. $b = \Sigma^{-1}\sigma_H$ are the slopes from regressing r_H on asset returns.

Per dollar of the account, $w = x\,(W+H)/W$, so with $h = H/W$:

$$w \;=\; (1+h)\,\frac{1}{\gamma}\Sigma^{-1}\mu \;-\; h\,b$$

- With a job of $\pm 0.5\times$ Tech, b is ± 0.5 in Tech and zero elsewhere. Add the job back: $\frac{W}{W+H}\,w + \phi\,b = x^\star$, the star.
- The job is valued as if it can be priced like the assets it moves with

Back

A: Data, and what to be careful about

- **Six factors.** Ken French, 1980–2026, excess returns. They are long-short, so "weights add to one" only sets a scale.
- **Eight sectors.** Ken French's 49 industries, grouped by us, equal-weighted within a sector, over T-bills, 1980–2026. "Other" is left out. "Sin" is Smoke, Guns and Beer. The benchmark is the sector mix that best tracks the market.
- **In sample.** The frontier uses the same data it is judged on. Its 94% in Sin is what happened, not a recommendation.
- **Human capital.** Every number is stylized: the job's beta on one sector is an assumption, chosen to make the tilt easy to see.
- **No costs, no capital limits.** Everything here is gross.